

SmartCare v0.2 - Requirements Specification Template

1. Problem and Scope

SmartCare currently is using spreadsheets, paper records and manual note taking to manage patients and appointments. This has produced difficulty in locating patient records, keeping track of current appointments and produced limited visibility of practitioner availability, along with many other issues.

The first version of the system will provide simple management of patient, practitioner and appointment information. It will support booking and cancellation, appointment status management, practitioner availability, patient searching and basic operational reporting. The system will remain a manageable clinic appointment system rather than a complex hospital information system.

2. Stakeholders

Stakeholder	Need	Evidence
Patients	Accurate information of their appointment and of their details.	Case study identifies problems with appointments and patient records.
Practitioners	View scheduled appointments and availability	Case study identifies limited visibility of practitioner availability.
Reception staff	Create, search, update and cancel patient and appointment information	Processes are identified but the exact reasonabilities are unknown.
Clinic manager	Reliable appointment information and report generation	Case study states management wants a system and basic reporting
System manager	Maintain a reliable and maintainable system	Requirement for a simple, maintainable application

3. Functional Requirements

FR-01: The system will allow staff to create a new patient record

FR-02: Staff will be able to search for a patient by name

FR-03: Staff will be able to update patient information

FR-04: Staff will be able to create an appointment for a patient

FR-05: Appointments won't be created when the selected practitioner is already booked for that time

FR-06: Staff will have the ability to view the practitioners schedule and availability

FR-07: Staff will be able to cancel an appointment

FR-08: Staff have the ability to update the status of the appointment

FR-09: Basic operational reports will be generated based on patient data

FR-10: Cancelled appointments, the booking slot should open up and able to be booked

FR-11: Patient, staff or practitioner information should be maintained

FR-12: The system shall allow staff to retrieve stored patient and appointment information

4. Non-Functional Requirements

NFR-01: The system shall maintain appointment and patient data without crashing during operation.

NFR-02: Staff will be able to complete simple tasks like searching a patient name without having the technical knowledge.

NFR-03: The system should be consistent with the patient, practitioner, staff and with appointment information so that an appointment cannot be associated with a non-existent patient or practitioner.

NFR-04: Individual components can be modified or maintained without crashing or changes.

NFR-05: The system should respond when operating with course-scale datasets.

NFR-06: Core business logic shall be independently testable without requiring the entire application to run.

5. User Stories

US-01: As a staff member, I want to create a patient record, so that stored and retrieved when needed.

US-02: As a staff member, I want to book an appointment, so that the patient can be seen by a practitioner.

US-03: As a staff member, I want to cancel an appointment, so that appointment information is up to date.

US-04: As a practitioner, I want to see my booking schedule, so that I can prepare for my appointments.

US-05: As a staff member, I want to update patient details, so that their information is up to date.

US-06: As a patient, I want my data to be stored safely, so that I can trust the clinic and feel protected.

6. Acceptance Criteria

GIVEN: A patient record exists

WHEN: a staff member enters the correct patient name

THEN: the matching patient record is displayed

GIVEN: a practitioner is already booked at a selected time

WHEN: a staff member attempts to create another appointment for that practitioner at the same time

THEN: the system rejects the booking and indicates that the practitioner is unavailable

GIVEN an existing appointment is selected

WHEN a staff member cancels the appointment

THEN the appointment status changes to cancelled and the appointment remains in the appointment history.

7. Assumptions and Open Questions

Assumptions

- The first version is intended to be a simple appointment-management system rather than a full hospital information system.
- Staff are the users responsible for the core administrative functions unless otherwise specified.
- Cancelled appointments should remain available as historical records.
- Appointment booking must prevent a practitioner from being double-booked.

Open Questions

- Should patient records be searched by ID, name, or both?
- What information must be stored in a patient record?
- What specific information should practitioner availability display?
- What operational reports are required?
- Which staff roles can create, update and cancel appointments?
- Can patients book or cancel their own appointments?
- What specific criteria should be used to measure usability?
- What level of modularity and maintainability does the client expect?

8. AI Requirements Review Record

AI suggestion	Evidence?	Decision	Reason	Verification
Make patient search method consistent	Requirements use both patient ID and patient name	Modified	The inconsistency could cause confusion between the requirement and acceptance criteria.	Confirm with client whether search should use ID, name, or both.
Define information stored in a patient record	FR-01 requires patient records but does not specify their contents	Unverified	The required information cannot be safely assumed.	Ask client what patient information must be stored.
Prevent duplicate practitioner bookings	FR-05 explicitly requires this	Accepted	Directly supported by the functional requirements and addresses a documented SmartCare problem.	Test by attempting to book two appointments for the same practitioner at the same time.
Clarify practitioner availability	FR-06 requires staff to view practitioner availability but does not define what this means	Unverified	“Availability” could have multiple interpretations	Ask client what availability information must be displayed.
Define basic operational reports	FR-09 requires basic reports but does not specify their contents	Unverified	The required reports cannot be invented without client evidence.	Ask client which operational reports are required.